A Fine-Tuned Small Language Model for Interpreting Fluid–Structure Interaction Experiments

A U-Net surrogate stands in for the solver and the language model reads what comes out — the whole digital twin runs on one laptop

***Kumar, Angad**
angadkumar16ak@gmail.com
Obra Tompkins High School, Texas, USA

Shukla, Khemraj
khemraj_shukla@brown.edu
Brown University, Rhode Island, USA

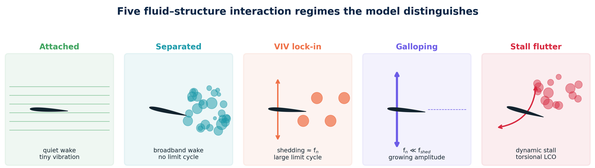
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CODE & DATA
github.com/AngadKumar16/FSI-SLM
REST API · MCP server · weights

1,928 solver cases • **107** airfoil sections • **R² = 0.998** lift, unseen shapes • **2.3%** velocity-field error • **0.86** lock-in F1 score • **1** laptop, no GPU

MOTIVATION

Fluid–structure interaction (FSI) decides when a hydrofoil or a turbine blade shakes itself apart. Reading what it produces — particle image velocimetry (PIV) fields, force and accelerometer traces — takes an expert, and **the expert is the bottleneck**. A frontier model talks about the physics fluently, invents what it does not know, and needs a data centre to do it.



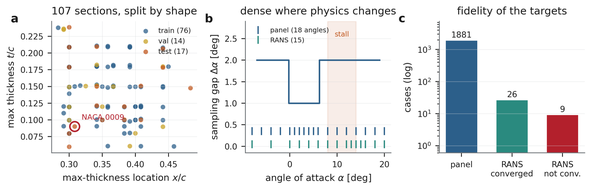
F1. The five flow–vibration regimes the interpreter must tell apart — the expert judgement being automated.

OBJECTIVES

1. Reduce any FSI experiment, measured or simulated, to **one machine-readable JSON record**.
2. Fine-tune a **small** language model to read the physics and flag likely sensor noise.
3. **Ground** each answer in published fluid dynamics by retrieval, so the model cites rather than invents.
4. Predict fields, forces **and** structural response from geometry, replacing the solver.
5. Serve all of it from an ordinary laptop over REST and MCP.

DATA

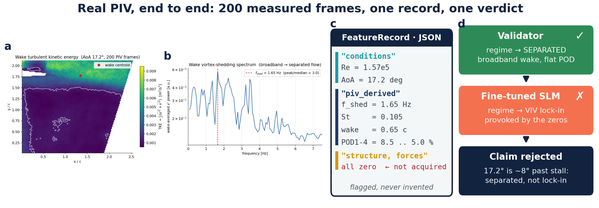
107 NACA sections, 18 angles of attack each at Re = 5×10⁵ — a 1,926-case design grid, **1,928 solved cases** on disk (1,881 panel, 35 RANS, 12 URANS). Three solvers write **one common tensor contract**, chosen by incidence: Hess–Smith panel with Beddoes–Leishman stall correction at any angle, steady RANS below stall, URANS above. A chordwise Euler–Bernoulli beam turns surface pressure into load and displacement, so every case carries structure as well as flow. **NACA 0009 is held out entirely.**



*F2. (a) Splits hold out whole **shapes**, so NACA 0009 (circled) is pinned to test and every number reported for it is a generalisation result. (b) Where the 18 angles per section sit. (c) Target fidelity, log scale; the 12 URANS cases are not plotted.*

A REAL PIV EXPERIMENT

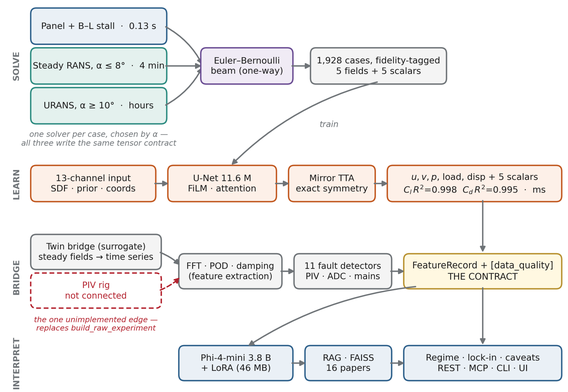
200 DaVis PIV frames of a NACA 0009 at **α = 17.2°**, 15 Hz on a 152 × 127 grid, through the **unchanged** feature extractor: measured and simulated share one code path.



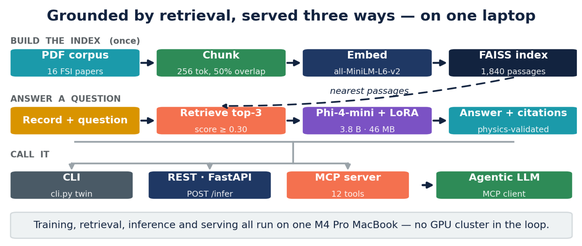
*F3. Channels a PIV-only export cannot carry are written as zero and **flagged**, never invented — and those zeros provoke an over-confident lock-in call, which the deterministic validator **overrules**. A broad wake and a flat POD spectrum are separated flow, not lock-in.*

METHODS

A 13-channel raster carries the geometry, including the **inviscid panel solution on the same grid**, so the 11.59 M U-Net predicts the **correction** to potential flow, not the flow. α, Re and fidelity enter by FiLM; attention sits at the 8×8 bottleneck because circulation is global. The reader is **Phi-4-mini** (3.8 B) plus a 46 MB LoRA adapter.



*F4. The twin end to end. The bridge turns a steady field into a time-resolved record, and eleven fault detectors label dropout, clipping and mains hum before the model sees anything. **The dashed red edge is the one unimplemented link;** F3 is the measured path running beside it.*



*F5. Retrieval-augmented generation (RAG) over **1,840 passages** puts the top three into the prompt, so an answer carries what it leaned on. A CLI, a **FastAPI REST** endpoint and a **Model Context Protocol (MCP)** server of twelve tools all reach the same pipeline — which is what lets an agent drive the twin.*

ANALOGY: WHY MULTI-FIDELITY

A panel solve costs 0.13 s at any angle; a steady RANS solve costs four minutes and only converges below stall; a URANS case costs hours. Train on the cheap tier alone and you get a fluent approximation with a **structural blind spot**, because the beam amplifies the leading-edge pressure difference — exactly where the solvers disagree. The cheap tier buys coverage; the expensive tier buys truth. The same argument sets the model size: a frontier model is a library with no lab notebook.

REFERENCES

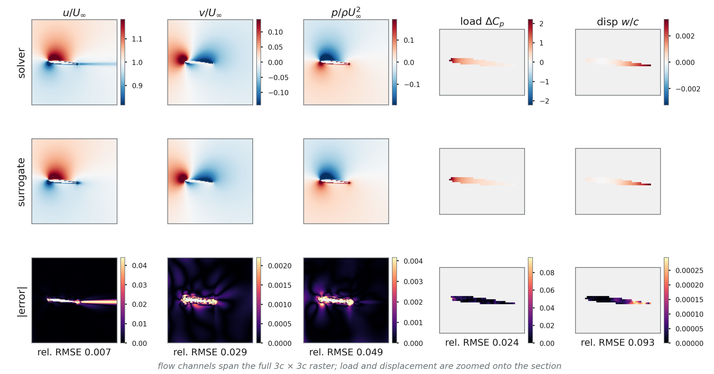
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ACKNOWLEDGEMENT

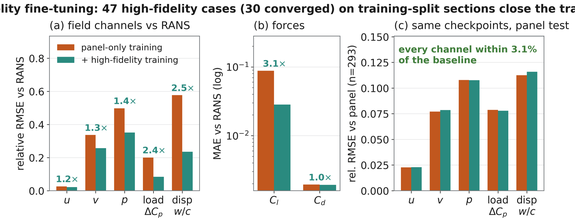
With thanks to the AGU Bright STaRS program. Built with Python, PyTorch, OpenFOAM, MLX and FAISS. Code, data and weights: **github.com/AngadKumar16/FSI-SLM**

RESULTS

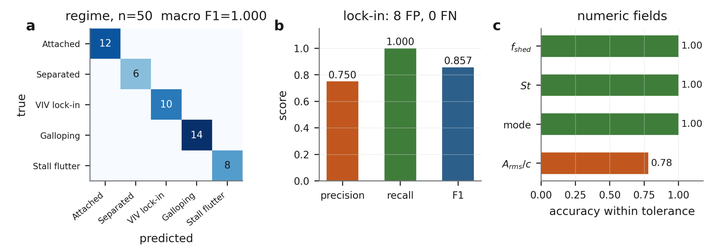
NACA 0009 (held out) at α = 5°, Re = 5×10⁵



*F6. Solver, surrogate and error on NACA 0009 at α = 5°, a section held out entirely. Relative RMSE 0.007 (u), 0.029 (v), 0.049 (p), 0.024 (load), 0.093 (displacement); error concentrates in the near-wall layer and the wake. Across **293 held-out cases on 17 unseen sections**, scored against the panel targets it was trained on: R^2 = 0.998 lift, 0.995 drag, 0.991 moment, 0.968 peak deflection, 0.786 trailing-edge twist.*



*F7. Panel-only training reproduces RANS velocity and drag but not the structural channels. Adding 47 high-fidelity cases on **training-split** sections closes most of that gap while every **field** channel stays within 3.1% of the panel baseline. The honest cost: panel-test drag MAE regresses 0.00230 → 0.00314; lift MAE improves 11%.*



*F8. The interpreter on 50 held-out records. Regime classification is saturated at this sample size — a fact about the test set as much as the model — so the discriminating metric is lock-in detection: recall 1.000, precision 0.750. It **never misses a lock-in** and over-calls eight — a bias the rule-based labeller shares, so it was inherited, not invented.*

SUMMARY

The sensor half predicts velocity, pressure, load and displacement fields plus lift, drag and moment on shapes it has never seen. The interpreter half reads the resulting JSON record, names the regime, flags sensor faults and cites the literature it used — and, on the real PIV run, is overruled when it over-reaches. Both halves run on one laptop, over REST and MCP.

FUTURE WORK

Acquire the structural channel alongside PIV so the real records carry displacement, not zeros. Finish the URANS sweep so the deep-stall branch becomes trainable. Close the structural loop so deflection feeds back into the flow. Add calibrated uncertainty on every field, and a harder interpreter test set now that regime accuracy has saturated.